

A Functionally-Grounded Benchmark Framework for XAI Methods: Insights and Foundations from a Systematic Literature Review

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Artificial Intelligence (AI) is transforming industries, offering new opportunities to manage and enhance innovation. However, these advancements bring significant challenges for scientists and businesses, with one of the most critical being the ‘trustworthiness’ of AI systems. A key requirement of trustworthiness is *transparency*, closely linked to *explicability*. Consequently, the exponential growth of eXplainable AI (XAI) has led to the development of numerous methods and metrics for explainability. Nevertheless, this has resulted in a lack of standardized and formal definitions for fundamental XAI properties (e.g., what do soundness, completeness, and faithfulness of an explanation entail? How is the stability of an XAI method defined?). This lack of consensus makes it difficult for XAI practitioners to establish a shared foundation, thereby impeding the effective benchmarking of XAI methods. This survey article addresses these challenges with two primary objectives. First, it systematically reviews and categorizes XAI properties, distinguishing them between human-centered (relying on empirical studies involving explainees) or functionally-grounded (quantitative metrics independent of explainees). Second, it expands this analysis by introducing a hierarchically structured, functionally grounded benchmark framework for XAI methods, providing formal definitions of XAI properties. The framework’s practicality is demonstrated by applying it to two widely used methods: LIME and SHAP.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence; Machine learning approaches;**
• **General and reference** → **Surveys and overviews; Evaluation; Metrics;**

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1 Introduction

Artificial Intelligence (AI) and **Machine Learning (ML)** have the potential to transform all industries, having already demonstrated impressive outcomes in many sectors such as health-care, genomics, agriculture, finance, Industry 4.0, social media, or forensics [1–3]. Despite the opportunities offered by AI/ML technologies, they are not yet fully accepted by the society due to misconceptions and misinformation in societal debates (leading to mistrust) [4], challenges of auditability (e.g., impossibility to access code/data due to corporate secrecy) [5, 6], or lack of transparency (difficulty for explainees to understand how and why the explainer reaches a given finding/decision) [7].

The lack of transparency in AI is a timely scientific challenge and a subject of increasing attention in the literature [9]. The complexity of ‘explaining’ the outcomes of an AI/ML model depends on its nature, which can be declined into two categories: (i) interpretable (aka. white-box) models such as linear regression, logistic regression, decision trees, which make it possible to understand the relationship between the model’s input data and output decisions; (ii) non-interpretable (aka. black-box) models such as **convolutional neural networks (CNNs)**, **graph neural networks (GNN)** or transformers, which outperform the more interpretable models in many tasks but suffer from a lack of insight of why and how the model acquires its findings, but also of what is being modelled for gaining a complete understanding of those findings. In recent years, a number of **eXplainable AI (XAI)** methods have been proposed to solve the lack of explainability in ML models, particularly when faced with black-box models [10]. According to the Ethics Guidelines for Trustworthy AI published by the European Union’s High Level Expert Group (AI HLEG) [8], *Explainability* is one requirement out of three that contribute to make AI systems Transparent,¹ the two others being *Traceability*: how data and labels have been gathered, what classifier has been used, and so on; and *Communication*: how explanations can be adapted to the explainee depending on his/her social situation (e.g., explanation of Task *i*’s outcome is communicated to John and Bob in a graphical way in Figure 1, which appears appropriate for John but not for Bob). Although *Explainability* can be implemented independently of traceability mechanisms, *Communication* is a pre-requisite of XAI, and is often considered as a part of the XAI concept.

An exponential number of XAI methods have been proposed in the past few years, which has also resulted in a growing number of survey articles. Nonetheless, as argued in [11–13], the majority of these surveys focus on categorizing XAI methods based on their functional properties, and too few perform experimental/numerical performance analyses. One reason for this comes from the difficulty of evaluating XAI methods, as it implies conducting (i) *human-centered evaluations*: empirical studies with the explainees [7], as the quality of an explanation is partly subjective (cf., example of John and Bob in Figure 1), and (ii) *functionally-grounded evaluations*: quantitative evaluation not involving explainees [14]. To date, most studies conducting performance analyses of

¹Transparency is listed as one requirement for Trustworthy AI among seven; the six remaining being (i) Accountability, (ii) Human agency and oversight, (iii) Technical Robustness and safety, (iv) Privacy and data governance, (v) Diversity, non-discrimination and fairness, (vi) Societal and environmental wellbeing.

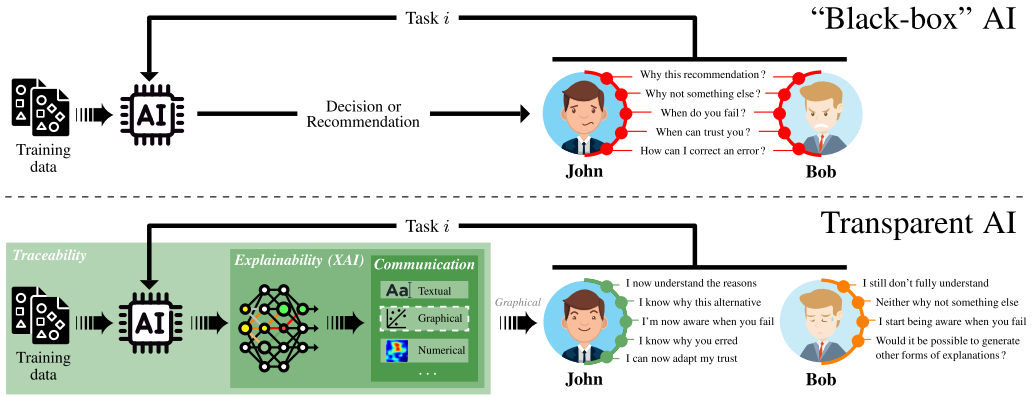


Fig. 1. Traditional “black-box” AI model vs. Transparent (XAI) model, as defined by the Ethics Guidelines for Trustworthy AI [8], which outlines Transparency through Traceability, Explainability, and Communication.

XAI methods are of human-grounded nature [15, 16]. Furthermore, due to the infancy and rapid growth of XAI research, there is still a lack of consensus of the properties underlying XAI, such as: What do soundness, completeness, and faithfulness of an explanation mean? What does stability of an XAI method signify? and so on. Even when it comes to the XAI concepts themselves (i.e., explainability, interpretability, transparency), there is no consensual definition [7, 17]. This goes without saying that it is difficult for AI practitioners to work on a common ground and perform benchmarking studies of XAI methods [18].

This survey article has two main objectives. First, after introducing key XAI concepts and providing a classification of existing XAI methods, it reviews and analyzes—in Section 2—XAI properties introduced in the literature, categorizing them by their nature (human-centered or functionally-grounded). Second, it builds on this review by proposing—in Section 3—a consensual, hierarchically structured functionally-grounded benchmark framework that formally defines XAI properties to the greatest extent possible. The framework’s practicality is demonstrated in Section 4 through an application involving two widely known and commonly used state-of-the-art XAI methods (LIME and SHAP). Section 5 discusses future research directions, while Section 6 provides the conclusions.

2 A Systematic Literature Review Approach

Section 2.1 details the methodology for the systematic review, while Sections 2.2 to 2.4 examine how XAI is defined, approached, and evaluated in the literature, respectively.

2.1 Methodology

To identify relevant scientific articles in the field of XAI, a search was carried out from 2015² to 2024 using the following keywords: {Explainable AI, Interpretable AI, Explainability ML, Interpretability ML}. The search was performed in six library databases: *Science Direct*, *arXiv*, *IEEE*, *Springer*, *Elsevier*, and *ACM*. Items such as doctoral dissertations, master’s theses and unpublished articles were ignored. As shown in Figure 2, 139 scientific articles were identified. After screening titles, abstracts, and excluding unfit studies, 95 articles were selected for the final

²Explainability in AI was an inherent requirement already in the MYCIN system in the 1970’s [19], discussed in the mid-1980s [20] and actively researched in the 1990’s [21], but it re-emerged around 2015 [22], when the term XAI was introduced by DARPA (Defence Advanced Research Project Agency).

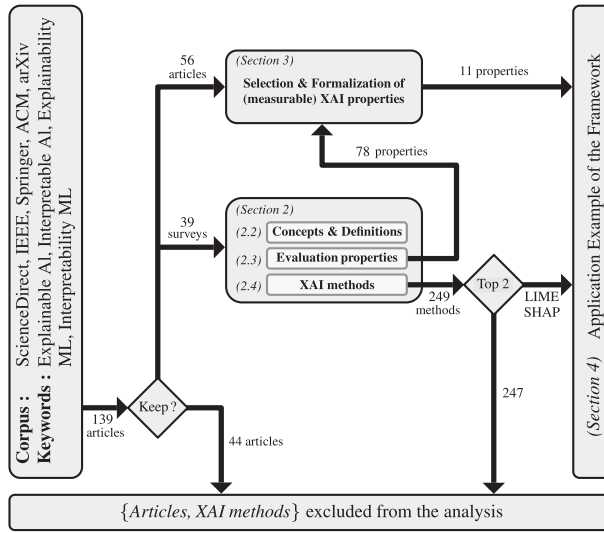


Fig. 2. Research methodology applied in the present article.

review stage, 39 being categorized as literature reviews [1–7, 9, 11–13, 17, 18, 23–48]. The research methodology followed in the present article to create the benchmarking framework is divided into distinct sections, first consisting—in *Section 2*—in analyzing the 39 systematic literature review articles to identify (i) what XAI methods exist at the moment and what problem(s) they apply to, and (ii) what properties are considered and defined in the literature to characterize and evaluate XAI methods. In *Section 3*, based on the findings of our analysis, we select, adapt, or propose a consensual, hierarchically structured functionally-grounded benchmark framework that formally defines XAI properties to the greatest extent possible.

2.2 XAI: Concepts and Definitions

After analyzing the 39 systematic literature review articles, we can argue that XAI—*despite its fast emerging*—is not yet a well-established field, demonstrating a lack of common formalism and taxonomy. As an evidence of this, one may find in the 39 survey articles different definitions for explainability, along with various concepts that sometimes overlap with each other, such as interpretability, transparency, intelligibility, comprehensibility, or still understandability. As of today, there is an ongoing debate about whether interpretability and explainability should be used interchangeably, as confirmed by our analysis (26% of the reviewed surveys use these terms interchangeably, 33% consider them to be different, and 41% do not discuss the matter). This underscores the need for more formalism in defining and quantifying these terms within the XAI literature [2, 9]. In the present article, we align with the definitions given by Linardatos et al. [34] for distinguishing interpretability and explainability, the former being the ability to discern the internal mechanics without necessarily knowing why, being mostly connected with the intuition behind the outputs of a model; the latter being the ability to explain what is happening (i.e., the system’s reasoning). In our opinion, explainability adds a reasoning layer on top of mechanics used to interpret ML models/systems. Regarding transparency, we adopt the definition of the Ethics Guidelines for Trustworthy AI [8], which states that transparency is achieved by fulfilling three requirements: Traceability, Explainability, and Communication, as illustrated in Figure 1.

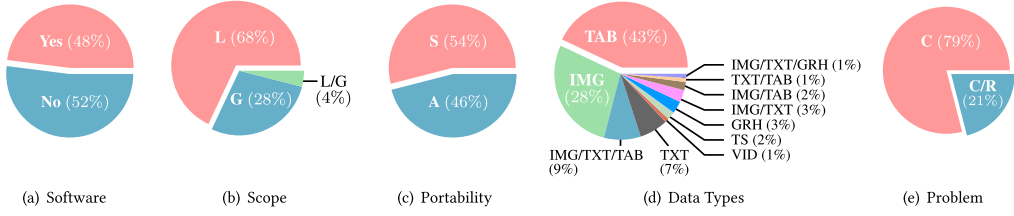


Fig. 3. Graphical summary of Table 1 with regard to each criterion (i.e., table’s column). A: model-agnostic; S: model-specific; G: global; L: local; TAB: tabular; IMG: images; TXT: text; TS: time series; VID: videos; GRH: graph; C: classification; R: regression.

2.3 XAI Methods

The complexity of an AI/ML model has a significant impact on the difficulty to interpret and explain the model [27], which often leads to a tradeoff between accuracy and explainability. To tackle this tradeoff dilemma, two explainability strategies are usually applied: (i) *Intrinsic methods (a.k.a., explainable-by-design methods)*: explainability is directly achieved through constraints imposed on the model during training (e.g., white-box models are intrinsically explainable); (ii) *Post-hoc methods*: applied to provide black-box explanations after model training [12, 33], thus avoiding the explainability vs. accuracy tradeoff. According to [27]: “as long as the model is accurate for the task, and uses a reasonably restricted number of internal components, intrinsic interpretable models are sufficient. If otherwise the prediction target involved complex and highly accurate models, considering post-hoc interpretation models is necessary”. Although some researchers use intrinsic methods for high-stake decisions [49], the literature reveals a strong trend favoring post-hoc approaches. Consequently, this article focuses exclusively on post-hoc methods.

From the 39 reviewed state-of-the-art surveys, a total of 249 post-hoc XAI methods were identified, which are reported in Table 1. The first three columns provide the *Name*, *Reference*, and publication *Year* of each method. The methods are sorted by the fourth column (% surveys), which indicates their popularity, calculated as the percentage of the 39 reviewed surveys in which each method is mentioned. The remaining columns detail:

- *Portability*: indicates the range of ML models to which the explanation method applies, distinguishing between (A) *model-agnostic methods*: used to explain any type of model, treating all ML models as black-boxes, even if they are not; (S) *model-specific methods*: used to explain only a specific type of black-box models;
- *Scope*: indicates the extent to which the method addresses the entire model behavior, distinguishing between (G) *global methods*: used to explain the entire model behavior (i.e., the overall logic of the black-box model), which means that a global explanation is valid for any instance; (L) *local methods*: used to explain the reasons for a single prediction, being valid for a specific instance;
- *Data*: indicates what types of data the method applies to, considering six data types: (TAB): tabular data; (IMG): images; (TXT): text; (TS): time series (VID): videos; (GRH): graph data;
- *Problem*: indicates to what kind of problems/tasks the method applies to, which can be (C) *Classification*: to categorize data into a given number of classes; and/or (R) *Regression*: to predict values for one variable given values from another;
- *Software (Soft.)*: indicates if the source code for reproducing the explanations is available: Yes (Y) or No (N).

To enhance the readability of Table 1, a graphical summary is provided in Figure 3. In Figure 3(a), one may observe that a significant portion of XAI methods (52%) lack publicly available source

Table 1. List of Post-hoc XAI Methods Identified through Our Systematic Literature Review

Name of XAI method	Ref	Year	% surveys	Portability	Scope	Data	Problem	Soft.
LIME	[50]	2016	76	A	L	TAB, IMG, TXT	C, R	Y
SHAP	[51]	2017	64	A	L	TAB, IMG, TXT	C, R	Y
Shapley values	[52]	2014	58	A	L	TAB	C, R	Y
LRP	[53]	2015	51	S	L	IMG, TXT	C	Y
Saliency Maps	[54]	2013	48	S	L	IMG, TXT	C	Y
Grad-CAM	[55]	2019	46	S	L	IMG	C	Y
IntGrad	[56]	2017	43	S	L	IMG, TXT	C	Y
Anchors	[57]	2018	41	A	L	TAB, TXT	C	Y
DeepLIFT	[58]	2017	38	S	L	IMG, TXT	C	Y
Influence Functions	[59]	2017	35	A	L	IMG	C	Y
TCAV	[60]	2017	35	A	G	IMG	C	Y
ICE	[51]	2015	35	A	G	TAB	C, R	Y
DTD	[61]	2017	33	S	L	IMG	C	Y
Local gradients	[62]	2010	33	A	L	IMG, TAB	C	N
DeconvNet	[63]	2013	33	S	L	IMG	C	Y
SmoothGrad	[64]	2017	30	S	L	IMG	C	Y
PDP	[65]	2001	30	A	G	TAB	C, R	Y
PatternAttribution	[66]	2017	28	S	L	IMG	C	Y
Guided BackProp	[67]	2014	28	S	L	IMG	C	Y
Meaningful Perturbation	[68]	2017	28	S	L	IMG	C	Y
PatternNet	[66]	2017	25	S	L	IMG	C	Y
Show, Attend and Tell	[69]	2015	20	S	L	IMG	C	Y
Activation Maximization	[70]	2010	20	S	L	IMG	C	Y
GSA	[71]	2011	20	A	G	TAB	C, R	N
DGN	[72]	2016	20	S	G	IMG	C	N
CAM	[73]	2015	20	S	L	IMG	C	Y
Counterfactuals	[74]	2017	20	A	L	IMG, TAB	C	N
Rationales	[75]	2016	17	S	L	TXT	C	Y
LORE	[76]	2018	17	A	L	TAB	C	Y
DeepRED	[77]	2016	17	S	L	IMG	C	N
PDA	[78]	2017	15	S	L	IMG	C	Y
RISE	[79]	2018	15	S	L	IMG	C	Y
CEM	[80]	2018	15	S	L	ANY	C	Y
Guided Proto	[81]	2019	12	A	L	IMG, TAB	C	Y
DICE	[82]	2020	12	A	L	TAB	C	Y
ACE	[83]	2019	12	A	G	IMG	C	Y
ConceptSHAP	[84]	2020	12	A	G	IMG	C	Y
Soft DT	[85]	2017	12	S	G	IMG	C	Y
ALE	[86]	2016	12	A	G	TAB	C, R	Y
TSP	[87]	2016	10	S	L	TAB	C	N
CIU	[88]	1995	10	A	L/G	IMG, TAB	C, R	Y
STA	[89]	2016	10	S	G	TAB	C	N
CaCE	[90]	2019	10	A	G	IMG	C	N
Regularisation	[91]	2015	10	S	L	IMG	C	Y
GFA	[92]	2016	10	A	G	TAB	C, R	Y
DT Extraction	[93]	2018	10	S	G	IMG	C	N
GoldenEye	[94]	2014	10	A	G	TAB	C	N
GAN Dissection	[95]	2018	10	S	L	IMG	C	Y
Discriminative Loss	[96]	2016	10	S	L	IMG	C	N
FACE	[97]	2019	10	A	L	ANY	C	Y
RxREN	[98]	2012	10	S	G	TAB	C, R	N
Distillation	[99]	2015	10	A	G	IMG	C	N
Feature Tweaking	[100]	2017	10	S	L	TAB	C	N
TREPAN	[101]	1996	10	S	G	TAB	C	N
SPRAY	[102]	2019	7	S	G	IMG	C	N
NAM	[103]	2020	7	S	L	TAB	C, R	Y
RuleMatrix	[104]	2018	7	A	G	TAB	C	Y
Grad-CAM++	[105]	2018	7	S	L	IMG	C	Y
L2X	[106]	2018	7	A	L	IMG, TXT	C	Y
MAPLE	[107]	2018	7	A	L	TAB	C, R	Y
PIMP	[108]	2010	7	A	G	TAB	C	Y
BreakDown	[109]	2018	7	A	L	TAB	C, R	Y
ABELE	[110]	2020	7	A	L	IMG	C	Y
PALM	[111]	2017	7	A	G	ANY	C, R	N
TreeView	[112]	2016	7	S	G	IMG	C	N
TreeEnsemble explainer	[113]	2016	7	S	G	TAB	C, R	N
inTrees	[114]	2014	7	S	G	TAB	C, R	N

(Continued)

Table 1. Continued

XAI method	Ref	Year	% surveys	Portability	Scope	Data	Problem	Soft.
MOC	[115]	2020	7	A	L	TAB	C, R	Y
Counterfactual SHAP	[116]	2019	7	A	L	ANY	C, R	N
Agnostic PFI	[117]	2018	7	A	G	TAB	C	N
MMD-critic	[118]	2016	7	S	G	TAB, IMG, TXT	C	N
SEDC	[119]	2014	7	A	L	TXT	C	Y
LRP*	[120]	2019	5	S	L	ANY	C	N
BiLRP	[121]	2020	5	S	L	ANY	C	N
LIVE	[109]	2018	5	A	L	TAB	C, R	Y
Erasure	[122]	2016	5	S	L	TXT	C	N
GFI	[123]	2018	5	S	L	IMG	C	N
MUSE	[124]	2019	5	A	L	IMG	C	N
Multifaceted feature visualizat*	[125]	2016	5	A	L	IMG	C	N
GraphLIME	[126]	2020	5	A	L	GRH	C	Y
GLocalX	[127]	2020	5	A	L/G	TAB	C	Y
Privacy-Preserving Explanations	[128]	2020	5	S	L	TAB	C	Y
Bayesian Teaching	[129]	2021	5	A	L	TAB	C, R	N
Network dissection	[130]	2017	5	S	L	IMG	C	N
PI, ICI	[131]	2018	5	A	G	TAB	C, R	Y
Distill-and-Compare	[132]	2018	5	A	G	TAB	C, R	N
Model Extraction	[133]	2017	5	A	G	TAB	C, R	N
Explanatory Graph	[134]	2017	5	S	G	IMG	C	N
Saliency Detection	[135]	2017	5	A	L	IMG	C	N
Compositionality	[136]	2016	5	S	L	TXT	C	N
ASTRID	[137]	2017	5	A	L	TAB	C	N
QII	[138]	2016	5	A	G	TAB	C	Y
RSRS Detection	[139]	2012	5	S	L	TXT	C	N
Worst-case perturbations	[140]	2015	5	A	L	IMG	C	N
Counterfactual Introspection	[141]	2019	5	S	L	IMG	C	N
CADEX	[142]	2019	5	S	L	TAB	C	N
Rule Extraction	[143]	1994	5	S	L	TAB	C	N
DT Extractor	[144]	1999	5	S	L	TAB	C	N
Concept Tree	[145]	2019	5	A	L/G	TAB	C	N
G-REX	[146]	2008	5	S	L	TAB	C, R	N
REFNE	[147]	2003	5	S	L	TAB	C	N
OAE	[148]	2015	5	S	G	TAB	C	N
EMAP	[149]	2019	5	A	G	TAB	C	N
MACEM	[150]	2019	5	A	G	TAB, IMG, TXT	C	N
CRUDS	[151]	2020	5	S	G	TAB	C	N
REVISE	[152]	2019	5	S	G	TAB	C	N
FOCUS	[153]	2021	5	S	G	TAB	C, R	Y
EBCF	[154]	2020	5	S	L	TAB	C	Y
DCE	[155]	2019	5	S	L	TAB	C	Y
Actionable Recourse	[156]	2019	5	S	L	TAB	C	Y
DACE	[157]	2020	5	S	L	TAB	C	Y
MACE	[158]	2020	5	A	L	TAB	C, R	Y
C-CHVAE	[159]	2020	5	A	L	TAB	C	Y
SYNTH	[160]	2020	5	S	L	TAB	C	Y
Recourse Summaries	[161]	2020	5	A	G	TAB	C	N
SCOUT	[162]	2020	5	S	L	IMG	C	N
FRACE	[163]	2020	5	S	L	IMG	C	N
DECE	[164]	2020	5	S	L/G	TAB	C	Y
ALG-REC	[165]	2020	5	S	L	TAB	C	Y
GIC	[166]	2017	5	S	L	TAB	C, R	N
Growing Spheres	[167]	2017	5	A	L	TAB	C	Y
POLARIS	[168]	2018	5	A	G	TAB, IMG, TXT	C, R	N
CF Visual Expl.	[169]	2019	5	A	L	IMG	C	N
CERTIFAI	[170]	2020	5	A	L	TAB, IMG, TXT	C, R	Y
Plausible CFs	[171]	2020	5	A	L	TAB	C, R	N
ViCE	[172]	2020	5	A	L	TXT	C	Y
PermuteAttack	[173]	2020	5	A	L	TAB	C	N
GRACON	[174]	2022	5	A	L	TAB	C, R	N
MC-BRP	[175]	2020	5	A	L	TAB	C, R	Y
LIME-C/SHAP-C	[176]	2020	5	A	L	TAB	C	Y
CLEAR	[177]	2019	5	A	L	TAB	C, R	Y
FOILTREE	[178]	2018	5	A	L	TAB, TXT	C	N
“Good” CFEs	[179]	2020	5	A	L	TAB	C, R	N
XGNN	[180]	2020	2	S	G	GRH	C	N
GNNE explainer	[181]	2019	2	A	L	GRH	C	Y

(Continued)

Table 1. Continued

XAI method	Ref	Year	% surveys	Portability	Scope	Data	Problem	Soft.
GNN-LRP	[182]	2022	2	S	L	IMG, TXT, GRH	C	N
LSTM-LRP	[183]	2019	2	S	L	IMG, TXT	C	N
On-manifold Shapley	[184]	2020	2	A	L	IMG, TXT, TAB	C,R	N
Shapley Flow	[185]	2020	2	A	L,G	TAB	C,R	Y
FBT	[186]	2020	2	S	G	TAB	C	Y
CHIRPS	[187]	2020	2	S	L	TAB	C	Y
ExMatrix	[188]	2021	2	S	L	TAB	C	N
LoRMika	[189]	2019	2	A	L	TAB	C	Y
NLP-PIPELINE	[190]	2021	2	A	L	TXT	C	N
BK-SVM	[191]	2019	2	S	G	TAB	C	N
XSVMC	[192]	2020	2	S	L	IMG	C	N
Color based monogram	[192]	2016	2	S	G	TAB	C	Y
SLRP	[192]	2021	2	S	L	IMG	C	N
CEFEs	[193]	2021	2	S	L	TS	C	N
SR map	[194]	2019	2	S	L	IMG	C	Y
DLIME	[195]	2019	2	A	L	ANY	C, R	Y
LioNets	[196]	2019	2	S	L	TXT	C	Y
SkopeRules	N/A	2020	2	A	L/G	TAB	C	Y
CFX	[197]	2020	2	S	L	TAB	C	N
XRAI	[198]	2019	2	S	L	IMG	C	Y
Respond-CAM	[199]	2018	2	S	L	IMG	C	N
XSPELLS	[200]	2020	2	A	L	TXT	C	Y
exBERT	[201]	2019	2	S	L	TXT	C	Y
Slot Activation Vectors	[202]	2018	2	S	L	TXT	C	Y
Peak Response	[203]	2018	2	S	L	IMG	C	Y
Autofocus-Layer	[204]	2018	2	S	L	IMG	C	Y
DNN-CLEAR	[205]	2017	2	S	L	IMG	C	N
DFA	[206]	2018	2	S	L	TAB	C	N
OpenBox	[207]	2018	2	S	G	IMG	C	N
Probes	[208]	2016	2	S	G	IMG	C	N
Relevant Features	[209]	2017	2	S	L	IMG	C	N
OPIA	[210]	2016	2	A	G	TAB	C, R	N
NNKX	[211]	2017	2	S	G	TAB	C	Y
Automatic Rule Extraction	[212]	2017	2	S	G	TXT	C	N
RxNCM	[213]	2017	2	S	G	TAB	C	N
MES	[214]	2016	2	A	L	ANY	C	N
Hypothesis Testing	[215]	2019	2	A	L	IMG, TXT	C	Y
GAM	[216]	2019	2	A	G	IMG, TAB	C	Y
Shapley Values-based	[217]	2010	2	A	L	TAB	C	N
Monotone Influence	[218]	2018	2	A	L	IMG	C	N
Rivelo	[219]	2017	2	A	L	TXT	C	N
SQB	[220]	2018	2	A	L	IMG, TAB	C	N
Textual CFs	[221]	2018	2	A	L	IMG	C	N
InterpNET	[222]	2017	2	A	L	IMG	C	Y
Important Neurons and Patches	[223]	2017	2	S	G	IMG	C	Y
Pertubation-based	[224]	2017	2	A	L	TXT	C	N
CNN-INTE	[225]	2018	2	S	G	IMG	C	N
Hidden Activity Visualization	[226]	2017	2	S	G	IMG	C	N
Causal Inference	[227]	2018	2	S	G	IMG	C	N
Rivelo	[219]	2017	2	A	L	TXT	C	N
SQB	[220]	2018	2	A	L	IMG, TAB	C	N
Textual CFs	[221]	2018	2	A	L	IMG	C	N
InterpNET	[222]	2017	2	A	L	IMG	C	Y
Important Neurons and Patches	[223]	2017	2	S	G	IMG	C	Y
Pertubation-based	[224]	2017	2	A	L	TXT	C	N
CNN-INTE	[225]	2018	2	S	G	IMG	C	N
Hidden Activity Visualization	[226]	2017	2	S	G	IMG	C	N
Causal Inference	[227]	2018	2	S	G	IMG	C	N
Autoencoded Activations	[228]	2018	2	S	G	IMG	C	N
ACE	[229]	2019	2	S	G	ANY	C	Y
LASTS	[230]	2020	2	A	L	TS	C	N
DoctorXAI	[231]	2020	2	A	L	TAB	C	Y
Multimodal Information	[232]	2018	2	S	G	VID	C	N
PFI	[233]	2001	2	S	G	TAB	C	N
Prototype Selection	[234]	2011	2	S	G	TAB, IMG, TXT	C	N
IBD	[235]	2018	2	S	G	IMG	C	N
AI Rationalization	[236]	2017	2	S	G	TXT	C	N

(Continued)

Table 1. Continued

XAI method	Ref	Year	% surveys	Portability	Scope	Data	Problem	Soft.
Propositional Rule Extract*	[237]	2017	2	S	L	TAB	C	N
CEODT	[238]	2021	2	S	L	TAB, IMG, TXT	C	N
SGNCE	[239]	2021	2	S	L	TAB	C	N
OCEAN	[240]	2021	2	S	L	TAB	C	N
ORDCE	[241]	2021	2	A	L	TAB	C	Y
PIECE	[242]	2021	2	S	L	IMG	C	Y
POLYJUICE	[243]	2021	2	A	L	TXT	C, R	Y
GRACE	[244]	2020	2	A	L	TAB	C, R	N
PCIG	[245]	2020	2	A	L	TAB	C	N
GeCO	[246]	2021	2	A	L	TAB	C, R	Y
NICE	[247]	2022	2	A	L	TAB, TXT	C	N
RF-OCSE	[248]	2020	2	A	L	TAB	C, R	Y
Weighted CFs	[249]	2018	2	A	G	TAB	C	N
What-If	[250]	2020	2	A	G	TAB, IMG, TXT	C, R	Y
PRINCE	[251]	2020	2	S	L	GRH	C	Y
SEDC-T	[252]	2020	2	A	L	IMG	C	N
ASP-based CFs	[253]	2020	2	A	L	TAB	C	N
Survival Model CFs	[254]	2020	2	A	L	TAB	C, R	N
Gradient Boosted CFs	[255]	2020	2	S	L	TAB	C	Y
TS CFs	[256]	2021	2	A	L	TS	C	N
PARE	[257]	2022	2	S	L	TAB	C	N
CluReFI	[258]	2020	2	A	L	TAB	C, R	N
CRED	[259]	2001	2	S	G	TAB	C	N
SA	[260]	2016	2	S	G	TXT	C	N
Score-CAM	[261]	2019	2	S	L	IMG	C	Y
ACV	[262]	2022	2	A	L	TAB	C, R	Y
BETA	[263]	2017	2	A	L	TAB	C	N
SoundLIME	[264]	2017	2	A	L	TS	C	Y
CMM	[265]	1998	2	A	L	TAB	C	N
Oracle guide	[266]	2009	2	A	L	TAB	C	N
ANOVA decomposition	[267]	2004	2	A	L	TAB	C, R	N
Prospector	[268]	2016	2	A	L	TAB	C, R	N
Visualization	[269]	2008	2	A	L	TAB	C	N
DarkSight	[270]	2018	2	A	L	IMG	C	N
Forest Floor	[271]	2016	2	S	G	TAB	C, R	Y
VQA explainer	[272]	2018	2	S	L	IMG	C	N
SQRex-SVM	[273]	2007	2	G	L	TAB	C	N
Symbolic interpr.	[274]	2002	2	G	L	TAB	C	N
Contribution Propagation	[275]	2013	2	S	L	IMG	C	N
ChemHeatmap	[276]	2011	2	S	L	GRH	C	Y
Nonograms	[277]	2004	2	S	L	TAB	C	N
DkNN	[278]	2018	2	S	L	IMG	C	N
Cocox	[279]	2020	2	S	L	IMG	C	Y
Inverting Method	[280]	2015	2	S	L	IMG	C	N
NAFER	[281]	2024	2	A	L/G	TAB	C	Y
Improved LRP**	[282]	2022	0	S	L	IMG, TXT, GRH	C	Y
"Better" CFs	[283]	2021	0	A	L	TAB	C	N
TRUST	[284]	2023	0	A	L	TAB	C	N
LIME&SHAP	[285]	2022	0	A	L	IMG, TXT, TAB	C, R	N
CIE	[286]	2021	0	A	L	TXT, TAB	C	Y
Class Prototypes	[287]	2021	0	A	L	IMG, TAB	C	N
Informative Features	[288]	2022	0	A	L	GRH	C	N
Score-CAM GNN	[289]	2022	0	S	L	GRH	C	N
XAI-GCNN	[290]	2019	0	S	L	GRH	C	N
LeGrad	[291]	2024	0	A	L	IMG	C	Y
SyntaxShap	[292]	2024	0	S	L	TAB	C	Y

Ref: reference; A: model-agnostic; S: model-specific; G: global; L: local; TAB: tabular; IMG: images; TXT: text; TS: time series; VID: videos; GRH: graph; C: classification; R: regression; Soft: software; Y: yes; N: no.

Grey highlighting indicates methods that were not cited in the reviewed articles but were identified through additional research by the authors.

code, suggesting that many are still at the proof-of-concept stage. Regarding the *scope* of applicability (Figure 3(b)), local explanations, which focus on individual predictions, dominate the field (68%). While ideally, XAI methods should support both local and global explanations, the complexity of achieving global explanations likely contributes to this trend [38]. For portability (Figure 3(c)),

Table 2. Link between Data Types and Explanation Types

Explanation type		TAB	TXT	IMG	VID	TS	GRH
Feature Summary	Explanations provide summary values (e.g., importance score or weight) for each feature / time point / time (sub)sequence / node / edge (commonly along with visualization).	✓				✓	✓
Rules	Explanations are presented in the form of condition rules (antecedent) that an instance must satisfy in order to meet the (predefined) consequent.	✓				✓	
Attention Map	Explanations are presented in the form of a matrix of scores that reveals how each feature/ word in the input table / text are related to each other (e.g., interaction score).	✓	✓				
Saliency Map	Explanations highlight the contribution of each pixel / word / video frame / node / edge to a particular prediction.			✓	✓	✓	✓
Concept Attribution	Explanations rely on the computation of attribution scores to predefined target “concepts” in the image/video. Explanations show how sensitive is the output (e.g., prediction of wolf) to a concept (e.g., presence of snow).			✓	✓		
Data points	Explanations can be of type (i) <i>prototypes</i> : data instances characterizing the predicted outcome; or (ii) <i>counter-factuals</i> : examples similar to the input instance, that are found by making the smallest change to some feature values that changes the prediction to a desirable output.	✓	✓	✓	✓	✓	✓

Adapted from [10, 33, 294–296].

TAB: tabular; TXT: text; IMG: images; VID: video; TS: time series; GRH: graph.

most methods are model-specific (54%), which may be influenced by the prevalence of image data (Figure 3(d)) and the popularity of CNN-specific methods for image recognition tasks. Nevertheless, scholars often favor model-agnostic methods due to their flexibility in handling various models, explanations, and representations [293].

Figure 3(d) shows that XAI methods for tabular data (TAB) are well-studied, unlike those for text (TXT), despite the large **Natural Language Processing (NLP)** research community [33]. This gap is even more pronounced for time series (TS), video (V), and graph (GRH) data, which are more complex to handle. Guidotti et al. [18] noted that until 2018, XAI research was limited to TAB, IMG, and TXT data. Another key insight from Figure 3(e) is that most XAI methods focus on classification problems, while regression (R) methods are primarily developed for TAB data ($\approx 50\%$). Additionally, the types of explanations generated by XAI methods are closely tied to the input data type [10, 33]. To complement Table 1 and its analysis, Table 2 summarizes the common types of explanations associated with various data types.³

Our review reveals a substantial number of methods proposed over the past decade, with the majority (71%) published between 2016 and 2020, as illustrated in Figure 4. The decline in the publication of new XAI methods from 2021 onward might be partly explained by a shift in research focus. Our interpretation is that, given the high number of XAI methods, scholars recognized the importance of categorizing and evaluating these methods, driven by the need to identify their *strengths* and *limitations*. Secondly, the fact that the number of AI applications in the industry worldwide has been doubling every year since 2018 points towards a hypothesis that scholars have increasingly focused on applying XAI methods in real-world use cases, rather than prioritizing the creation of new ones.

2.4 Evaluation Properties

Evaluating an XAI method is not a straightforward process, at least not as simple as evaluating the performance of an ML model, due to the difficulty, sometimes impossibility to access ground-truth labels/explanations. What makes it even more challenging is that the quality of an explanation is partly subjective, as it depends on how each explainee perceives the explanation (cf., Figure 1). This is why XAI methods are evaluated either through empirical studies conducted with the explainees [7] or through the use of quantitative measures that do not require the involvement of explainees. These two approaches are referred to as human-grounded for the former, and functionally-grounded for the latter [14].

³Note that some XAI methods produce explanations that do not fit neatly into the categories in Table 2.

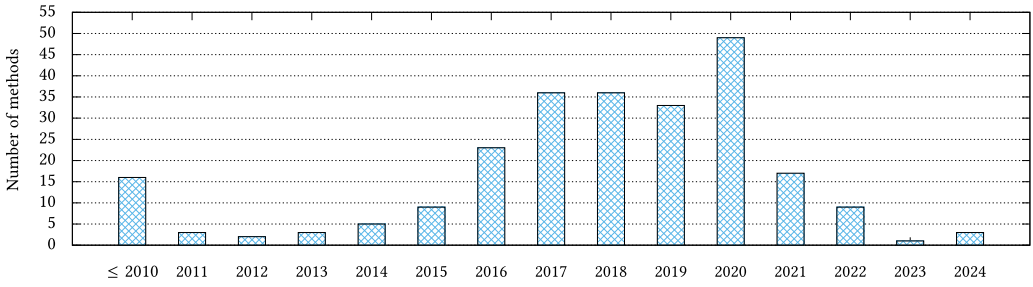


Fig. 4. Trend of XAI method creation/publication over the years.

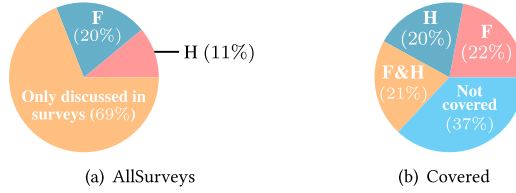


Fig. 5. Graphical summary of Table 3 (columns S, F, H).

Various properties for evaluating XAI methods have been proposed in the literature. Table 3 provides an overview of the 78 identified properties based on an analysis of 39 survey articles. The column labeled “S” indicates the number of surveys (out of 39) that introduce or define a property. Columns “H” and “F” report the number of surveys proposing human-centered (H) or functionally-grounded (F) property formalizations, respectively. The final column includes a brief *Description* of each property. Figure 5(a) illustrates the proportion of surveys that merely introduce the 78 properties (without formalization) versus those offering human-grounded or functionally-grounded definitions. A complementary pie chart in Figure 5(b) highlights the proportion of these properties that are formally defined (human-grounded, functionally-grounded, or both) versus those lacking formalization. Notably, a substantial share of surveys (69%) only discuss properties without providing definitions or mathematical formalizations. This analysis underscores the need for more functionally-grounded metrics, which complement human-grounded approaches [35]. Other researchers, such as Burkart and Huber [36], emphasize the importance of establishing “a standard procedure to measure, quantify, and compare the explainability of different approaches, enabling scientists to evaluate them consistently”. Additionally, many of the identified properties overlap significantly, with similar concepts labeled differently across surveys. This lack of consensus, as noted in [2], complicates the comparison of XAI methods and highlights the necessity for standardized definitions and formalizations to ensure consistent evaluations.

Despite the above-described situation, some studies have recently attempted to address this gap. Sokol and Flach [297] introduced a taxonomy that comprises five dimensions. However, their approach primarily offers a qualitative assessment, lacking computational and mathematical formulations. Similarly, Cui et al. [298] presented an integrated framework emphasizing correlation, completeness, and complexity. While they provide a holistic understanding, their evaluations remain qualitative. Alikhademi et al. [299] directed their efforts towards identifying potential pitfalls in XAI, focusing on issues like fairwashing. Their qualitative framework centers on issues related to biased data, model selection and creation, and XAI result presentation. From a quantitative perspective, the QUANTUS toolbox [300] provides a diverse set of metrics that encompass six categories: (1) faithfulness, (2) robustness, (3) localisation, (4) complexity, (5)

Table 3. SoTa Property Identification

Property	S	F	H	Description
P1 Interpretability	19	1	6	Whether the human can gain understanding into the model's decision strategy [25].
P2 Fidelity	17	4	0	To which extent a surrogate model is able to accurately mimic a black-box predictor [33].
P3 Trustworthiness	13	0	8	Confidence of whether a model will act as intended when facing a given problem [13].
P4 Accuracy	13	3	1	It is related to the predictive accuracy of the explanation regarding unseen data [11].
P5 Causa(bi)lity	12	0	1	Capacity of a method to clarify the relationship between input and output [31].
P6 Comprehensibility	11	0	4	It is related to how well humans understand the explanations [11].
P7 (In)Stability	11	3	0	How (in)consistent the explanations are for similar instances with the same class label [17].
P8 Faithfulness	10	3	1	Checks if the meaningful features predicted by the model are really meaningful [36].
P9 (Explanation) complexity	9	2	3	Degree of effort required by a user to comprehend an explanation [33].
P10 Understandability	9	0	6	Capacity of a method of explainability to make a model understandable [31].
P11 Usability/Usefulness	9	0	6	Actual usability of explanations from the point of view of the end-user [33].
P12 Social	8	0	0	Explanations should be delivered as a conversation [29].
P13 (User) Efficiency	8	0	6	Capacity of a method for explainability to support faster user decision-making [35].
P14 (Un)certainty	8	0	1	Refers to providing a comprehensible (un)certainty measure and accounting for randomness in explanation generation [11].
P15 Robustness	8	2	0	The persistence to withstand small perturbations of the input data [31].
P16 Consistency	7	2	0	How deterministic and implementation-invariant the explanation method is [38].
P17 Soundness	6	1	0	Extent to which the explanation is truthful in describing an underlying system [35].
P18 Sensitivity	6	1	0	Ability to capture model sensitivity to input feature variations [31].
P19 Computat* complexity	6	1	0	It is related to computational/algorithmic complexity of the explanation method [11].
P20 Reliability	6	0	2	Ability to maintain performance regardless of parameters or input data [18].
P21 (Computational) Cost	6	0	0	Addresses the time of the methods [33].
P22 Interactivity	6	0	0	Ability to interpret and respond to follow-up questions using prior context [35].
P23 Output-completeness	5	2	1	Completeness as measured on a substitute task [26].
P24 Sparsity/Size	5	1	1	Sparsity considering the limited cognitive capacity of humans [36].
P25 Confidence	5	1	2	Describes the presence and accuracy of probability information in the explanation [38].
P26 Selectivity	5	0	0	Explanation methods should be able to provide selected explanations [12].
P27 Contrastivity	5	1	1	Describes how discriminative the explanation is w.r.t. other events or targets [38].
P28 Expressive power	5	0	1	Language or structure of the explanations the method is able to generate [11].
P29 Correctness	4	2	1	Addresses the faithfulness of the explanation with respect to the model [38].
P30 Relevancy	4	0	2	If the explanation provides meaningful information for a specific audience [36].
P31 Coverage	4	0	0	The most important features should be visible in the representation [35].
P32 Model-Completeness	3	2	2	Completeness of the model compared to the original model [26].
P33 Speed	3	3	0	Whether explanations can be obtained quickly with finite compute resources [32].
P34 Transferability	3	0	0	Capacity of a method to transfer prior knowledge to unfamiliar situations [35].
P35 Compositionality	3	1	1	Describes the format and organization of the explanation [38].
P36 Bias Detection	3	2	1	Ability to detect models with biases [26].
P37 Completeness	3	0	0	Addresses the extent to which the explanation explains predictive model [38].
P38 Translucency	3	0	0	How much the method relies on looking into the inner workings of the model [11].
P39 Clarity	3	0	1	Assesses if the explanation is unambiguous [12].
P40 Satisfaction	3	0	2	Measures the direct content of the explainee with the system. [48].
P41 Portability	3	0	0	Describes the range of models to which the explanation method can be applied [11].
P42 Graphical integrity	2	0	0	Extent to which representations highlight key features and differentiate attributions [31].
P43 Morphological clarity	2	0	0	Addresses how the important features are displayed [31].
P44 Layer separation	2	0	0	Assesses whether the representation occludes the raw image [31].
P45 Separability	2	0	0	Data instances from different populations must have dissimilar explanations [30].
P46 Implem. constraints	2	0	0	Time and compute requirement of the explainable algorithm [12].
P47 Truthfulness	2	0	1	Good explanations are proven to be true in the real world [11].
P48 Similarity	2	0	0	Data instances close to each other should generate similar explanations [30].
P49 Novelty	2	0	1	If the position of the studied instance in the training data distribution is shown [11].
P50 Broadness	2	1	0	Assesses how generally applicable an explanation is [12].
P51 Fairness	2	0	0	The predictions should not include any implicit or explicit bias against targeted users [9].
P52 Mental model	2	0	2	Explanations should help users establish better mental models / system's mechanisms [41].
P53 Human-AI interface	1	0	1	Performance of the human in the given task [9].
P54 Continuity	1	1	1	Describes how continuous and generalizable the explanation function is [38].
P55 Identity	1	0	0	Identical data instances must produce identical attributions or explanations [30].
P56 Representativeness	1	0	0	It describes how many instances are covered by the explanation [11].
P57 Applicability	1	0	0	Range of models that the method can be applied to [32].
P58 Importance	1	0	0	How well the explanation reflects the importance of features [12].
P59 Compactness	1	1	0	Size of explanations, motivated by human cognitive capacity limitations [38].
P60 Parsimony	1	1	0	Assesses whether the explanation is presented in a simple and compact form [12].
P61 Proximity	1	1	0	Counterfactual explanations should be as similar as possible to the original instance [17].
P62 Controllability	1	1	1	Indicates to what extent a user can control, correct or interact with an explanation [38].
P63 Context	1	1	1	Describes how relevant the explanation is to the user and their needs [38].
P64 Coherence	1	1	1	Describes how accordant the explanation is with prior knowledge and beliefs [38].
P65 Post-Expl. Performance	1	0	0	Evaluate the explanations by comparing the model performance before and after adjustment [41].
P66 Localization	1	1	0	Evaluate the explanations through localization capability [41].
P67 Plausibility	1	1	0	Combination of features in the explanation are realistic with respect to existing examples [42].
P68 Discriminative power	1	1	0	How discriminative are the reasons for a decision outcome [42].
P69 Actionability	1	1	0	Changed features in a counterfactual should be only actionable features (that can be mutated) [42].
P70 Diversity	1	0	0	Counterfactual explanations should provide a diverse set of examples [42].
P71 Optimality	1	0	0	Counterfactual instances should be proximal to the factual instance [43].
P72 Access	1	0	0	How many levels of access to the underlying dataset or model are required [43].
P73 Generalizability	1	0	0	Evaluate the utility due to the diversity of model architectures [9].
P74 Specificity	1	0	0	Capability of indicating which explanations are the key reasons for a prediction [9].

(Continued)

Table 3. Continued

	Property	S	F	H	Description
P75	Sociological	1	0	0	The social context and intended audience should be considered [9].
P76	Abnormality	1	0	0	Capability to identify odd circumstances that may have an influence on the outcome [9].
P77	Factuality	1	0	0	Explanation is plausible and relevant to other examples' predictions [9].
P78	Scalability	1	0	0	The XAI method should be scalable to larger datasets [47].

S: number of surveys that introduce or mention the property. F: number of surveys that propose or mention functionally-grounded metrics. H: number of surveys that propose or mention human-grounded metrics.

Table 4. Functionally-grounded Properties for Evaluating XAI Methods Based on State-of-the-art Definitions and Formalizations

Functionally properties	Description	Developer	Deployers	End-user
F1 Representativeness	Describe what is being provided and to what extent	■	■	
F2 Structure	Describe how the explanation is provided			■
F3 Selectivity	Consider the size of the explanation			■
F4 Contrastivity	Assess how contrastive the explanation is with regard to some reference target	■		■
F5 Interactivity	Assess the extent to which a user can control or explore explanations		■	■
F6 Fidelity	Assess if the method creates a surrogate model or assumes linearity	■		
F7 Faithfulness	Describe how reliable the explanation is with regard to the black box	■		
F8 Truthfulness	Describe how reliable the explanation is with regard to the true world			■
F9 Stability	Assess how stable and consistent the method is	■	■	■
F10 (Un)certainty	Assess how transparent the provided information for explanation is	■	■	■
F11 Speed	Assess the computation time for obtaining the explanation			■

randomisation, and (6) axiomatic metrics. While it offers a thorough and specialized framework for quantitative assessment, it is primarily designed for analyzing image data and does not cover certain dimensions, such as Representativeness, Structure, or Fidelity of XAI methods. Additionally, Schwalbe and Finzel proposed a taxonomy of XAI methods [48] that accounts for notions present in the current state of research, including evaluation metrics. While their work offers valuable clarity and a general overview of key XAI topics, it does not provide mathematical formalizations and fails to cover important dimensions. Our framework aligns with [297] in considering the diverse needs of user groups and mirrors the QUANTUS tool's quantitative focus.

3 Functionally-grounded Property Formalization

The literature review in the previous section highlighted that efforts to define functionally-grounded properties are both limited and fragmented. To address this gap, we advance the state-of-the-art by proposing a concise, consensual, and hierarchically structured framework for functionally-grounded properties. The framework is described as “concise” because it consolidates overlapping properties; “consensual” as it builds, where possible, on established formalizations reviewed in Section 2.4; and “hierarchically structured” as it organizes the properties in a clear, layered manner. Additionally, aligning with key research challenges [2, 9, 301], our framework incorporates multidisciplinary perspectives and accommodates varying levels of user expertise, aiming to enhance the trustworthiness and practical utility of XAI.

Table 4 provides an overview of the functionally-grounded properties (labeled F1 to F11) proposed for inclusion in our XAI benchmark framework. For each property, we: (i) offer a brief textual description; (ii) identify the primary user groups most impacted by the property, based on the three target groups defined by the High-Level Expert Group [8]: *Developers*, responsible for implementing and applying XAI methods; *Deployers*, the organizations ensuring the AI system complies with legal requirements; and *End-users*, individuals using the AI system who require

Table 5. Overview of the Functionally-grounded Properties, Sub-properties, and Associated Metrics Incorporated to Our Framework

Property	Sub-property	Origin	Dependency	Metric
F1 - Representativeness	F1.1 - Scope	P56	Fixed	Equation (1.1)
	F1.2 - Portability	P37-38, P41, P57, P73	Fixed	Equation (1.2)
	F1.3 - Access	P72	Fixed	Equation (1.3)
	F1.4 - Practicality	P78	Fixed	Equation (1.4)
F2 - Structure	F2.1 - Expressive Power	P28, P35, P52, P75	Data Type, User	Equation (2.1)
	F2.2 - Graphical Integrity	P42, P68	Data Type	Equation (2.2)
	F2.3 - Morphological Clarity	P39, P42-43	Data Type	Equation (2.3)
	F2.4 - Layer Separation	P44	Data Type	Equation (2.4)
F3 - Selectivity		P24, P26, P59-60, P74	Dataset	Equation (3)
F4 - Contrastivity	F4.1 - Contrastivity Level		Data Type	Equation (4.1)
	F4.2 - Target Sensitivity	P27, P45, P76	BB, Dataset	Equation (4.2)
F5 - Interactivity	F5.1 - Interaction Level		Fixed	Equation (5.1)
	F5.2 - Controllability	P11-12, P22, P52, P63, P65	Fixed	Equation (5.2)
F6 - Fidelity	F6.1 - Fidelity Check		Fixed	Equation (6.1)
	F6.2 - Surrogate Agreement	P2, P4, P32	BB, Dataset	Equation (6.2)
F7 - Faithfulness	F7.1 - Incremental Deletion	P8, P17, P23, P29	BB, Dataset	Equation (7.1)
	F7.2 - ROAR	P8, P17, P29	BB, Dataset	Equation (7.2)
	F7.3 - White-Box Check	P29, P66	WB	Equation (7.3)
F8 - Truthfulness	F8.1 - Reality Check	P67, P69, P71	Fixed	Equation (8.1)
	F8.2 - Bias Detection	P51, P76	BB, Dataset	Equation (8.2)
F9 - Stability	F9.1 - Similarity	P7, P15, P18, P20, P48	BB, Dataset	Equation (9.1)
	F9.2 - Identity	P55	BB, Dataset	Equation (9.2)
F10 - (Un)Certainty		P14, P25, P49	Fixed	Equation (10)
F11 - Speed		P19, P21, P33, P46	BB, Dataset	Equation (11)

A cell highlighted in green means it is our proposal; highlighted in orange means it is an adaptation from an existing article; no highlight means it is directly derived from an existing article. BB: black-box model, WB: white-box model.

explanations for its outputs (e.g., data scientists, clinicians, patients). The 11 properties and their associated sub-properties/metrics are detailed in Sections 3.1 to 3.11, and summarized in Table 5. For each sub-property, we: (i) specify which of the 78 properties from Table 3 are used as inputs (i.e., sub-criteria/metrics); (ii) indicate whether the metric is fixed (case-independent) or dependent on factors such as data type, dataset, or black-box model; and (iii) provide a direct reference to its corresponding equation in the text.

3.1 Representativeness (F1)

Developers/deployers of AI systems have to make a design choice regarding what type of model and input data the XAI method must support, but also how deeply the XAI method goes in describing the internal workings of the black-box. This is what we refer to as *representativeness* of an XAI method. *Representativeness* is divided into four sub-properties denoted by F1.1 to F1.4, which are further discussed below and summarized in Table 5:

Scope (F1.1): This property evaluates whether the XAI method provides explanations for the entire model behavior (*Global* explanation) or for a single prediction (*Local* explanation), as discussed in Section 2.3. The metric is adapted from P56 [11], which is evidenced by the orange highlight in Table 5. The property name has been simplified, and the original two-level scale has been expanded to a three-level scale, as defined in (1.1), with higher scores indicating better performance. Based on scientific studies (e.g., [27, 302]), we claim that Local explanations are preferred over Global ones, which is the reason why XAI methods supporting Local explanations get a higher score.

$$m_{f1.1} = \begin{cases} 3 & \text{the XAI method supports both Global and Local explanations,} \\ 2 & \text{the XAI method is Local,} \\ 1 & \text{the XAI method is Global.} \end{cases} \quad (1.1)$$

Portability (F1.2): This property assesses the range of ML models to which the XAI method can be applied, as was discussed in Section 2.3. In the literature, portability is known to be inversely proportional to translucency [11], which represents how much the XAI method relies on the inner workings of the underlying model. Practically speaking, it means that *model-specific* methods are highly translucent but of low portability (e.g., GradCAM, LRP), while *model-agnostic* methods have low translucency but are highly portable (e.g., LIME, SHAP). This sub-property is derived from P41⁴ [11], and is transformed into a two-level scale, as shown in (1.2), with higher scores assigned to model-agnostic methods due to their enhanced generalizability (P73).

$$m_{f1.2} = \begin{cases} 2 & \text{the XAI method is model-agnostic,} \\ 1 & \text{the XAI method is model-specific.} \end{cases} \quad (1.2)$$

Access (F1.3): This property assesses the degree to which the XAI method requires access to the training data or the model itself to generate a (new) explanation. Based on P72, two complementary metrics are introduced: Data Access ($m_{f1.3A}$) and Model Access ($m_{f1.3B}$). Together, these metrics determine the overall privacy-preserving capability of the XAI method, as outlined in (1.3). This aspect is critical for practical applications due to ethical and legal considerations. The final score, ranging from 0 to 5 (most privacy-preserving), is obtained by summing these two components.

$$m_{f1.3} = m_{f1.3A} + m_{f1.3B} \quad (1.3)$$

Where:

$$m_{f1.3A} = \begin{cases} 3 & \text{no (training) data access required,} \\ 2 & \text{data required only for initialization} \\ & \text{(e.g., creating an object explainer),} \\ 1 & \text{data required for each explanation,} \\ 0 & \text{full dataset required for retraining.} \end{cases} \quad (1.3A)$$

$$m_{f1.3B} = \begin{cases} 3 & \text{no model required (any function } f(X)), \\ 2 & \text{prediction function access only,} \\ & \text{(i.e., queries to a trained model),} \\ 1 & \text{partial access} \\ & \text{(e.g., layer outputs, gradients),} \\ 0 & \text{full access to model internals.} \end{cases} \quad (1.3B)$$

Practicality (F1.4): This property assesses an XAI method's applicability across diverse data types and its efficiency in handling large datasets (scalability, P78). We introduce a two-component scoring system to assess these dimensions, $m_{f1.4A}$ for applicability and $m_{f1.4B}$ for scalability, as formalized in (1.4). Methods that are data-agnostic and highly scalable receive higher scores, with the final score reflecting their practical flexibility:

$$m_{f1.4} = m_{f1.4A} + m_{f1.4B} \quad (1.4)$$

Where:

$$m_{f1.4A} = \begin{cases} 2 & \text{data-agnostic} \\ & \text{(applicable to all data types),} \\ 1 & \text{partially data-specific} \\ & \text{(requires preprocessing/adjustments),} \\ 0 & \text{fully data-specific} \\ & \text{(limited to specific data types).} \end{cases} \quad (1.4A)$$

$$m_{f1.4B} = \begin{cases} 2 & \text{highly scalable} \\ & \text{(efficient for large datasets),} \\ 1 & \text{moderately scalable} \\ & \text{(some limitations but can be adjusted),} \\ 0 & \text{not scalable} \\ & \text{(impractical for large datasets).} \end{cases} \quad (1.4B)$$

⁴When other properties are listed in Table 5 (e.g., P37-38, 57, 73 in F1.2), it means that they are overlapping with the selected one (P41).

3.2 Structure (F2)

When designing an XAI system, it is essential to focus on how the explanation is presented to explainees—*maximizing clarity in visualization while minimizing noise and ambiguity*—rather than on what is explained (addressed under Representativeness) or how the explanation is generated (based on the system’s theoretical foundation). This property is closely tied to the “communication” requirement discussed in Section 1 and Figure 1. It is further divided into four sub-properties F2.1 to F2.4. While these properties are defined in a functionally grounded manner, it is worth noting that human-grounded evaluation—*beyond the scope of this article*—plays a critical role in assessing the overall structure.

Expressive power (F2.1): This property assesses the extent to which the XAI method supports diverse and comprehensible representation formats or languages. Adapted from P28 (cf. Table 5), this metric was transformed from qualitative to numerical, as defined in (2.1), where n is the number of distinct explanatory outputs (e.g., importance scores, contrastive scores), $|F|$ is the number of unique representation formats (textual summaries, visual graphs, etc.), C is a predefined set of comprehensible formats,⁵ including logical structures (e.g., decision trees or rules) [33], textual formats [29], conceptual representations (e.g., high-level overviews or causal diagrams) [38], visualizations (e.g., heatmaps and bar plots), and analogies or metaphors (e.g., relatable examples for lay users), and $\mathbb{1}(f \in C)$ is an indicator function that equals 1 if the format f belongs to the set C , and 0 otherwise. Methods that provide greater flexibility in explanations, measured by the number of explanatory outputs (n) and representation formats (F), receive higher scores, provided that these formats belong to a comprehensible set C . Notably, Expressive Power is not fixed; it depends on the data type (different types of data necessitate varying outputs/formats) and the user (comprehensibility varies across individuals).

$$m_{f2.1} = n + |F| + \frac{\sum_{f \in F} \mathbb{1}(f \in C)}{|F|}. \quad (2.1)$$

Graphical integrity (F2.2): This property assesses whether the explanation effectively distinguishes between positive and negative attributions. Adapted from [31] (P42), we propose a Boolean metric to assess this property:

$$m_{f2.2} = \begin{cases} 1 & \text{positive and negative attributions are visually distinguishable,} \\ 0 & \text{positive and negative attributions are not visually distinguishable.} \end{cases} \quad (2.2)$$

Morphological Clarity (F2.3): This property assesses whether the explanation effectively distinguishes more relevant attributions from less relevant ones in a visually prominent way (e.g., longer bars, more intense colors). As outlined in (2.3), we propose a Boolean metric to assess this property, adapted from [31] (P43).

$$m_{f2.3} = \begin{cases} 1 & \text{the most relevant attributions are visually distinguishable,} \\ 0 & \text{relevant attributions are not visually distinguishable from less relevant ones.} \end{cases} \quad (2.3)$$

Layer Separation (F2.4): This property assesses whether the explanation visualization omits or occludes the original input instance, which should be visible for explainee inspection [11]. As outlined in (2.4), we propose a Boolean metric to assess this property, adapted from [31] (P44).

$$m_{f2.4} = \begin{cases} 1 & \text{the original input instance is visible,} \\ 0 & \text{the original input instance is omitted or occluded.} \end{cases} \quad (2.4)$$

⁵Comprehensibility of formats can depend on user feedback, allowing C to be updated based on user studies or domain-specific guidelines. Importantly, C can be tailored to the target audience, allowing more complex or less complex formats depending on the end-user’s expertise and specific needs.

3.3 Selectivity (F3)

XAI methods should be able to provide selective explanations that are concise enough for the explainee to evaluate, with the level of conciseness varying by individual [10, 35]. This property, referred to as *selectivity*, emphasizes clarity without overwhelming the user. While some studies [12, 38] propose metrics for specific data types, a universal approach remains elusive. In this survey, we adopt *Miller's Law* [303]: “the number of items an average person can process and remember in short-term memory is 7 ± 2 ”. The proposed metric, detailed in (3), evaluates selectivity based on this principle. Explanation size (s)—*measured as the number of attributes, regions, or words* (cf., Table 1 Supplementary Material)—achieves the highest score when $s = 7$ or when the method allows adjustment to $s = 7$. For non-optimal values, the score decreases using a Gaussian decay function, with $\sigma = 2$ reflecting the standard deviation in Miller's Law.

$$m_{f3} = \begin{cases} 1 & \text{if } s = 7 \text{ or tunable to } s = 7, \\ \exp\left(-\frac{(s-7)^2}{2 \cdot \sigma^2}\right) & \text{if } s \neq 7 \text{ and not tunable.} \end{cases} \quad (3)$$

3.4 Contrastivity (F4)

Humans often think in counterfactual terms, asking questions like, “How would the prediction change if input X were different?” [7]. Therefore, it is important to evaluate how well an explanation distinguishes the given outcome from an alternative event or target [2, 38]. This concept, referred to as the *contrastivity* of an XAI method, can be further divided into two sub-properties, Contrastivity Level (F4.1) and Target Sensitivity (F4.2).

Contrastivity Level (F4.1): This property assesses how effectively the generated explanation provides contrastivity relative to a standard reference point (e.g., a predefined output such as the average prediction, as in the Shapley value, or a selected case/point, as in CIU). This aspect has not been formally addressed in existing XAI surveys, as indicated by the **green** highlight in Table 5. To assess this property, a three-level scoring system is proposed, as detailed in (4.1), with higher scores awarded to methods that explain the model output in relation to a greater number of targets/events.

$$m_{f4.1} = \begin{cases} 2 & \text{the explanation compares the output with multiple targets/events,} \\ 1 & \text{the explanation compares the output with a single reference point,} \\ 0 & \text{no contrastivity is provided.} \end{cases} \quad (4.1)$$

Target Sensitivity (F4.2): This property assesses the robustness of an XAI method against adversarial attacks. These involve minimally altering inputs to produce (significant) changes in the model's outputs [304], and a reliable explanation should accurately reflect these shifts [47]. Methods for generating adversarial samples differ based on data type: for tabular data, a nearest counterfactual approach can be applied; for images, pixel intensities can be selectively modified; and for text, key words can be substituted with synonyms. The metric, adapted from [38] (P27), is formalized in (4.2), where E_1 represents the explanation before adversarial perturbation, E_2 the explanation after perturbation, $d(E_1, E_2)$ the distance between the two explanations (e.g., Euclidean distance for feature contributions or structural similarity index for saliency maps, detailed in Table 2 Supplementary Material), and $d_{\max}$ the maximum distance for normalization.

$$m_{f4.2} = \frac{d(E_1, E_2)}{d_{\max}}. \quad (4.2)$$

3.5 Interactivity (F5)

Ideally, explanations should allow the explainer and the explainee to interact with each other, implying according to [27] that “*the explainer must be able to leverage the mental model of the explainee while engaging in the explanation process*”. Enabling AI system deployers to assess how interactive their system will be is crucial (what types of visualisation are supported, to what extent users can control or explore explanations taking his/her social context into consideration [29], etc.). Interactivity is divided into two sub-properties:

Interaction level (F5.1): This property assesses the degree to which the XAI method allows for user interaction, using a three-level scale:

$$m_{f5.1} = \begin{cases} 2 & \text{interactive control is provided by default (e.g., predefined interaction patterns),} \\ 1 & \text{no default interaction but interaction can be implemented (e.g., via APIs),} \\ 0 & \text{no interaction is provided, and creation of interaction is complex.} \end{cases} \quad (5.1)$$

Controllability (F5.2): This property assesses the extent to which the XAI method can be controlled and how it enhances explanation quality. This encompasses mechanisms such as enabling user feedback, refining explanations based on user input, or offering alternative examples. We propose adapting the metric identified by Nauta et al. [38] using a five-level evaluation scale, as defined in (5.2).

$$m_{f5.2} = \begin{cases} 4 & \text{full user control with dynamic feedback improving explanations,} \\ 3 & \text{partial control with significant user influence on explanations,} \\ 2 & \text{limited control with pre-defined options for refinement,} \\ 1 & \text{minimal control (e.g., only visual exploration),} \\ 0 & \text{no control over explanations.} \end{cases} \quad (5.2)$$

3.6 Fidelity (F6)

When implementing an XAI method, it is crucial for developers to assess whether the explanation is derived from (i) a surrogate model, (ii) linearity assumptions about the model, or (iii) neither of these. This property, known as fidelity, reflects the extent to which an explanation represents the true decision-making process of the underlying model. This property is divided into two sub-properties:

Fidelity check (F6.1): This property assesses whether the fidelity to the underlying model is preserved. Methods that do not rely on a surrogate model or linearity assumptions receive a higher score. This is because explainability is most critical for complex, non-linear models, where employing a linear surrogate may inherently misrepresent the model’s behavior, reducing the fidelity of the explanation. This metric is assessed using a Boolean score.

$$m_{f6.1} = \begin{cases} 1 & \text{no surrogate model or linearity assumptions are used,} \\ 0 & \text{a surrogate model or linearity assumptions are used.} \end{cases} \quad (6.1)$$

Surrogate Agreement (F6.1): This assesses—when a surrogate model is used—the extent to which its predictions align with those of the black-box model. The evaluation metric, derived from [33] (P2), involves computing the normalized complement of the average prediction difference between the surrogate model (s) and the black-box model (b). This is formalized in (6.2), where N represents the number of instances evaluated, $b(x_i)$ denotes the black-box model prediction for instance x_i , $s(x_i)$ corresponds to the surrogate model prediction for instance x_i , and $\max(b(x))$ refers to the maximum value of the black-box model predictions (1 for classification tasks). If no surrogate

model is used, the score is assigned the maximum value: $m_{f6.2} = 1$.

$$m_{f6.2} = 1 - \frac{\sum_{i=1}^N |b(x_i) - s(x_i)|}{N \cdot \max(b(x))}. \quad (6.2)$$

3.7 Faithfulness (F7)

It is crucial to assess the ability of an XAI method to reliably capture the behavior of the black-box model, either at a global or local level. This concept is known as faithfulness. In this article, we emphasize that fidelity and faithfulness are distinct properties and should be evaluated separately. Indeed, a developer can always construct an alternative model that replicates the original model's predictions (resulting in a high fidelity score, $m_{f6.2}$), yet still produce arbitrarily manipulated explanation maps, leading to low faithfulness [38]. Faithfulness is assessed based on three sub-properties:

Incremental Deletion (7.1): This property assesses how the progressive removal of input features identified as relevant by the XAI method impacts the predictive model's output f . It is specifically designed to assess local methods by analyzing feature-specific explanations at the instance level. We adopt the formalization from [38] (P29), as defined in (7.1), where N represents the number of evaluated instances, n denotes the number of incrementally removed features, $f_j() :$ is the black-box model prediction for the j^{th} instance, i corresponds to the i^{th} relevant feature according to the XAI method, and i_r represents the i^{th} feature removed based on a random explainer.

$$m_{f7.1} = \frac{1}{N} \sum_{j=1}^N \frac{\int_0^n (f_j(i_r) - f_j(i)), di}{\int_0^n f_j(i_r), di}. \quad (7.1)$$

This normalized metric measures the faithfulness of an XAI method by comparing the cumulative effect of removing features identified as relevant by the XAI method versus those selected randomly. This approach helps detect whether removing a feature reveals a "null attribute" (i.e., a feature whose omission has no impact on the model's output), which should correspond to an importance score of zero [38]. Rather than directly deleting features, we employ a perturbation-based strategy to ensure that their removal does not generate out-of-distribution samples. This prevents violations of a fundamental assumption in ML: *the training and evaluation data must come from the same distribution*. Readers interested in further details can refer to [15, 38].

RemOve And Retrain (ROAR) (7.2): This property assesses the impact of incrementally removing input features identified as relevant by the XAI method on model accuracy after retraining. This metric is primarily suited for global methods, as it assesses the overall influence of feature importance across the dataset. To quantify this, we estimate the area between the curves (comparing the XAI method to a random explainer), as defined in (7.2). Specifically, n represents the number of incrementally removed features, $a(i)$ denotes the model accuracy after retraining with the i^{th} feature removed by the XAI method, $a(i_r)$ corresponds to the model accuracy after retraining with the i^{th} feature removed by a random explainer, $e(i_r)$ indicates the model error after retraining with the i^{th} feature removed by the XAI method, and $e(i_r)$ represents the model error after retraining with the i^{th} feature removed by a random explainer.

$$m_{f7.2} = \begin{cases} \frac{\int_0^n (a(i_r) - a(i)), di}{\int_0^n a(i_r), di} & \text{for classification tasks,} \\ \frac{\int_0^n (e(i) - e(i_r)), di}{\int_0^n e(i_r), di} & \text{for regression tasks.} \end{cases} \quad (7.2)$$

For XAI methods that do not explicitly provide feature importance scores (e.g., Shapley values), alternative techniques can be used to infer feature relevance. Counterfactual explanations, for

instance, allow features to be ranked based on the magnitude of changes between the original and counterfactual instances. Moreover, the approach to determining feature importance varies by data type: for image data, it can be derived from pixel-level attributions or observed variations; for text data, importance can be assessed through token attributions or by analyzing modifications in keywords (e.g., replaced or removed tokens). These methods ensure that the Incremental Deletion (F7.1) and ROAR (F7.2) metrics remain applicable across different data types and XAI methods.

White-Box Check (7.3): This property assesses whether the explanation aligns with the actual reasoning of the underlying model, which can be assessed by training a white-box model to mimic a black-box one. This metric, adapted from [38] (P29), is defined in (7.3), relying on a four-level scale representing different degrees of agreement between the generated explanations and the “ground-truth” explanations. It is particularly well-suited for tabular data, where white-box models are more accessible. The method for measuring agreement depends on the explanation format. For feature summary approaches like LIME, a linear regression model serves effectively as a white-box, as its coefficients directly indicate feature influence [305]. Similarly, for rule-based explanations, a decision tree acts as an ideal white-box model, allowing a direct comparison between its decision pathways and the explanations generated by the XAI method.

$$m_{f7.3} = \begin{cases} 3 & \text{agreement} \geq 95\%, \\ 2 & 80\% \leq \text{agreement} < 95\%, \\ 1 & 60\% \leq \text{agreement} < 80\%, \\ 0 & \text{agreement} < 60\%. \end{cases} \quad (7.3)$$

3.8 Truthfulness (F8)

When using an XAI method, the correspondence between the explanation and the user’s true world should be taken into consideration, which includes being accordant with prior relevant domain knowledge and beliefs of the “explainee”, but also being able to detect biased models [26]. Truthfulness is divided into two sub-properties, Reality Check (8.1) and Bias Detection (8.2), which are further detailed hereinafter.

Reality Check (8.1): This property assesses whether the XAI method prevents the generation of unrealistic data samples, ensuring alignment with real-world constraints derived from training data and domain knowledge. This involves not only verifying that individual feature values adhere to predefined limits but also maintaining the observed correlations between features within generated explanations. Adapted from [42, 43] (P67, 69, 71), this metric is assessed across two complementary dimensions:

- (1) *Feature constraints consistency* ($m_{f8.1A}$): Ensures that generated explanations do not violate feature bounds. For instance:
 - Tabular Data: Features like income and age should remain within valid ranges.
 - Image Data: Saliency maps should not introduce artifacts and distortions outside the training domain.
 - Text Data: Token-level changes should preserve grammatical accuracy and semantic coherence.
- (2) *Feature Correlation Consistency* ($m_{f8.1B}$): Ensures that generated instances respect the observed correlations between features in the training data. For example:
 - Tabular Data: Highly correlated features should vary consistently.
 - Image Data: Explanations should avoid altering regions that strongly correlate with the target class.
 - Text Data: Explanations should maintain coherence with known patterns of word usage.

The final score $m_{f8.1}$ ranging from 0 (least realistic) to 2 (most realistic), is calculated as in (8.1).

$$m_{f8.1} = m_{f8.1A} + m_{f8.1B} \quad (8.1)$$

Where:

$$m_{f8.1A} = \begin{cases} 1 & \text{explanations respect feature constraints,} \\ 0 & \text{explanations violate feature constraints.} \end{cases} \quad (8.1A)$$

$$m_{f8.1B} = \begin{cases} 1 & \text{explanations respect feature correlations,} \\ 0 & \text{explanations violate feature correlations.} \end{cases} \quad (8.1B)$$

Bias Detection (8.2): This property evaluates whether the XAI method can reveal biases within the model or dataset, such as spurious correlations, feature imbalances, or demographic stereotypes. By identifying these biases, the XAI method goes beyond merely explaining model predictions, helping for debugging, enhancing fairness, and improving overall model reliability. The proposed metric, inspired by the concept of abnormality (P76) [9], is assessed as in (8.2). This metric can be implemented using known biased models or synthetic data to evaluate the XAI method's ability to expose biases. Examples include: (i) Husky-vs-Wolf Classifier [293]: a model biased toward associating wolves with snowy backgrounds; (ii) Gendered Occupation Models: Models reinforcing stereotypes, such as associating “nurse” with females or “engineer” with males; (iii) Simulated Bias (Synthetic Data): Train a deliberately biased model by introducing spurious correlations and assess whether the XAI method identifies the introduced bias.

$$m_{f8.2} = \begin{cases} 1 & \text{bias is successfully exposed by the XAI method,} \\ 0 & \text{bias is not detected or irrelevant features are highlighted.} \end{cases} \quad (8.2)$$

3.9 Stability (F9)

Developers must ensure that the XAI method is robust to (small) changes in the input data. This property is divided into two sub-properties:

Similarity (9.1): It evaluates the consistency of explanations for similar input samples (neighbors). A high similarity score is generally desirable, as similar data points (e.g., inputs with the same label but slightly different features) are expected to yield comparable explanations [11, 31]. To illustrate, “if one says that a second balcony increases the price of a house, then it should also apply to other houses, at least of similar type” [10]. The measurement of similarity has been widely discussed in the reviewed literature, with common approaches including the average SSIM between neighboring instances and the average similarity coefficient. The method for defining neighboring instances can vary based on the task and data type. For tabular data, an effective approach is outlined in Shapash [306]. The similarity metric is defined as in (9.1), where N denotes the number of evaluated instances, x_{ij} represents the explanation generated for the j^{th} neighbor of the i^{th} instance, n corresponds to the number of neighboring samples for each evaluated instance, $d(x_{1j}, x_{ij})$ measures the pairwise distance for explanations (e.g., Euclidean distance, SSIM, or cosine similarity),⁶ k corresponds to the number of components in the explanation, σ_l is the standard deviation of the l^{th} component across neighbors, and μ_l represents the mean of the l^{th} component across neighbors.

$$m_{f9.1} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{n} \sum_{j=1}^n \frac{1}{1+d(x_{1j}, x_{ij})} & \text{if pairwise distance is meaningful,} \\ 1 - \frac{1}{k} \sum_{l=1}^k \frac{\sigma_l}{\mu_l} & \text{otherwise.} \end{cases} \quad (9.1)$$

⁶Distances should be computed on normalized explanation values, either locally (within neighbors) or globally (across the dataset).

Identity (9.2): This property assesses, through multiple experiment repetitions, the degree to which the generated explanations remain consistent across different runs of the XAI method for the same instance, i.e., identical input values rather than similar ones as in F9.1. Higher variability indicates greater instability in the XAI method [17, 30]. We adapt the metric from F9.1, but instead of comparing explanations across neighboring instances, we assess the consistency of explanations for the same instance. The metric is defined in (9.2), where N denotes the number of evaluated instances, x_{ij} represents the explanation generated during the j^{th} run for the i^{th} instance, r corresponds to the number of repeated runs for the same instance, $d(x_{1j}, x_{ij})$ measures the pairwise distance for explanations (e.g., Euclidean distance, SSIM, or cosine similarity), k corresponds the number of components in the explanation, σ_l is the standard deviation of the l^{th} component across runs, and μ_l represents the mean of the l^{th} component across runs.

$$m_{f9.2} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{r} \sum_{j=1}^r \frac{1}{1+d(x_{1j}, x_{ij})} & \text{if pairwise distance is meaningful,} \\ 1 - \frac{1}{k} \sum_{l=1}^k \frac{\sigma_l}{\mu_l} & \text{otherwise.} \end{cases} \quad (9.2)$$

Let us note that the choice of the pairwise distance measure $d(x_{1j}, x_{ij})$ depends on the explanation type (cf., Table 2 Supplementary Material). For feature-based and example-based explanations, Euclidean distance can compare feature attributions. For saliency maps, structural measures like SSIM are more effective in capturing spatial similarities. The choice of the distance function is flexible, ensuring the metrics broad applicability across diverse XAI methods.

3.10 (Un)Certainty (F10)

Besides explanations, it is important to provide decision-makers with a level of confidence in the XAI method's output, which is what is referred to as *(un)certainty*. This property is defined in (10), whose score ranges from 0 (no confidence measures) to 5 (fully transparent confidence reporting).

$$m_{f10} = \sum_{i=1}^5 c_i. \quad (10)$$

Where c_i corresponds to a binary indicator for the i^{th} confidence aspect:

- $c_1 = 1$: Confidence in the black-box model's result is reported (e.g., by displaying the model accuracy [18]).
- $c_2 = 1$: Confidence in the XAI explanation is reported.
- $c_3 = 1$: Random processes in explanation generation are disclosed (e.g., sampling/random perturbation [11]).
- $c_4 = 1$: The instance's distance from the training data distribution is indicated.
- $c_5 = 1$: Additional confidence indicators/measures are provided.

3.11 Speed (F11)

This property assesses the computational time needed for an XAI method to generate an explanation, ensuring it remains practical for real-world applications [17]. The proposed metric, adapted from P33, is defined on a 5-level scale based on explanation runtime, where t denotes the explanation generation time in seconds. The time thresholds for each level are aligned with the three key response time limits [307], which reflect human perceptual processing capabilities.

$$m_{f11} = \begin{cases} 0 & t > 10\text{sec}, \\ 1 & 5 < t \leq 10\text{sec}, \\ 2 & 1 < t \leq 5\text{sec}, \\ 3 & 0.1 < t \leq 1\text{sec}, \\ 4 & t \leq 0.1\text{sec}. \end{cases} \quad (11)$$

Table 6. Values for the Functionally-grounded Properties Computed Based on Table 5 Regarding the Selected XAI Methods

Property	Sub-property	LIME	kernelSHAP	treeSHAP
F1 - Representativeness	F1.1 - Scope	2	2	3
	F1.2 - Portability	2	2	1
	F1.3 - Access	4	4	3
	F1.4 - Practicality	3	2	3
F2 - Structure	F2.1 - Expressive Power	4.7	3.0	4.3
	F2.2 - Graphical Integrity	1	1	1
	F2.3 - Morphological Clarity	1	1	1
	F2.4 - Layer Separation	1	1	1
F3 - Selectivity		1	1	1
F4 - Contrastivity	F4.1 - Contrastivity Level	1	1	1
	F4.2 - Target Sensitivity	0.1	0.2	0.2
F5 - Interactivity	F5.1 - Interaction Level	1	1	1
	F5.2 - Controllability	2	2	2
F6 - Fidelity	F6.1 - Fidelity Check	0	0	1
	F6.2 - Surrogate Agreement	0.7	0.6	1
F7 - Faithfulness	F7.1 - Incremental Deletion	0.4	-	1
	F7.2 - ROAR	0.5	-	2
	F7.3 - White-Box Check	0.5	0.1	2
F8 - Truthfulness	F8.1 - Reality Check	1	1	2
	F8.2 - Bias Detection	1	1	1
F9 - Stability	F9.1 - Similarity	0.2	0.2	0.2
	F9.2 - Identity	0.2	1.0	1.0
F10 - (Un)Certainty		2	2	2
F11 - Speed		3	2	4

4 Application Example of the Framework

To demonstrate the practical use of the proposed functionally-grounded benchmark framework, we apply it to two of the most widely recognized XAI methods identified in our survey study (see Table 1): LIME [50] and SHAP (divided into KernelSHAP and TreeSHAP) [51, 308]. For this purpose, we use the Pima Indian Diabetes dataset as a case study, with a Random Forest model serving as the black-box classifier. The complete evaluation process, along with the source code, has been made publicly available in our GitHub repository.⁷ The experiments were conducted using Python 3.10.10 on a Microsoft Windows 10 Pro system, equipped with a 2.3 GHz 8-Core Intel Core i7 processor.

Table 6 presents the scores obtained by each method for each property, with higher scores reflecting better performance in the respective property. To facilitate comparison, the scores were normalized and visualized using spider plots (see Figure 6). Normalization was applied using the observed minimum and maximum values for each (sub-)property. Examining Figure 6(a), it can be noted that the framework captures significant differences in performance among the evaluated methods, particularly in key properties such as fidelity (F6), faithfulness (F7), truthfulness (F8), stability (F9), and speed (F11). This comparison reveals that LIME stands out in practicality (F1.4) and speed (F11) but struggles with stability (F9), especially in identity (F9.2), where explanations

⁷<https://github.com/DCanha/FUNCXAI-11>, last accessed on May 13, 2025.

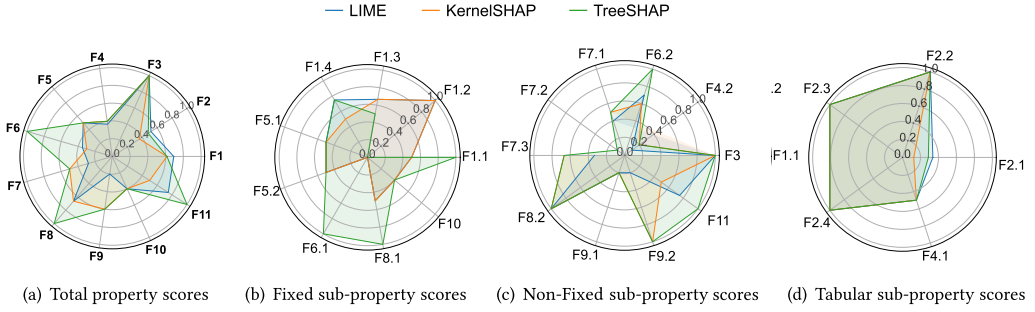


Fig. 6. Visualization of normalized scores (0-1) from Table 6.

vary across runs, raising concerns about consistency. It also exhibits low fidelity (F6), making it less reliable for complex models. Conversely, treeSHAP consistently outperforms in fidelity (F6), faithfulness (F7), and stability (F9), making it more suitable for high-stakes applications. However, its lower portability (F1.2) restricts its applicability to specific ML models. Meanwhile, kernelSHAP provides a well-balanced performance across most properties but does not excel in any particular area. Its higher computational cost (F11) may be a limiting factor, particularly in real-time applications. All methods show low scores on target sensitivity (F4.2), uncertainty estimation (F10), and interactivity (F5). The low scores on F4.2 and F10 suggest the need, respectively, for more robust methods capable of capturing nuanced model behavior, and more transparent and deterministic methods capable of assessing the confidence of their explanations. The low scores on F5 highlight the lack of interactivity in state-of-the-art XAI methods, which typically provide static, one-shot explanations without the ability to adapt to user feedback or background knowledge.

Fixed properties (e.g., F1 and F5) provide a consistent baseline for evaluating methods, ensuring comparability across diverse settings. Figure 6(b) shows that the most notable differences in fixed properties are in portability (F1.2), fidelity Check (F6.1) and reality Check (F8.1). LIME and KernelSHAP score higher in F1.2 due to their model-agnostic nature, while treeSHAP outperforms in F6.1 and F8.1, indicating stronger alignment with the underlying model without relying on surrogate models or assuming feature independence. Non-fixed properties (e.g., F7 and F9), whose results are displayed in Figure 6(c), emphasize variations arising from dataset and or/ model-specific factors. Our framework highlights that for the conducted experiments, TreeShap demonstrates superior performance in faithfulness (F7), scoring the highest across all three sub-properties (F7.1, F7.2, and F7.3), while also demonstrating superior stability (F9). Data type-dependent properties (e.g., F2), whose results are given in Figure 6(d), provide consistency within specific domains (in this case, the displayed values are for the tabular domain). Our framework shows no major differences in how the methods structure explanations (F2), except for Expression Power (F2.1), where LIME scores higher due to the rule-based format of each feature explanation (e.g., *Glucose* ≤ 100.00 decreases diabetes probability).

5 Future Directions and Research Opportunities

This systematic literature review establishes a concise, consensual, and hierarchically structured framework of functionally-grounded properties for benchmarking XAI methods. This framework encompasses tabular, image, text, time series, and graph input data, as identified in our review, and is applicable to regression and classification tasks.

It is designed as an initial assessment toolbox for system developers, researchers, and practitioners, helping them identify the most suitable XAI methods for specific applications. It can

serve as the foundation for advanced decision support tools, leveraging **multi-attribute decision-making (MADM)** methods such as AHP, MAUT, TOPSIS, and PROMETHEE. These tools would enable the combination and prioritization of various properties and sub-properties, such as emphasizing stability (F9) for consistency or speed (F11) for real-time applications. End-users of XAI methods could thus benefit from this framework and more advanced decision support tools for: (i) *method selection*: identifying the most suitable XAI method by assessing its ability to produce meaningful, actionable, and interpretable explanations; (ii) *model selection*: choosing the model that best aligns with domain knowledge and user expectations by evaluating how well its explanations reflect true behavior [309].

While the framework has proven effective in evaluating popular XAI methods like LIME and SHAP (see Section 4), further refinements are needed to address evolving challenges. A key improvement area is enhancing its adaptability to emerging interactive, social, and user-centered XAI paradigms. As highlighted by Miller et al. [310], a disconnect exists between how XAI researchers define explanations and what end users expect. Human-to-human explanation dynamics, extensively studied in social sciences, psychology, and decision sciences, remain largely unintegrated into XAI research. The **Social XAI (sXAI)** framework [311] addresses this gap by emphasizing: (i) *multimodality*: delivering explanations through diverse formats (text, graphs, gestures, etc.); (ii) *incrementality*: enabling co-constructive dialogues [312] where explanations are progressively refined; (iii) *partner models*: adapting explanations to user preferences, including vocabulary, question types, and modalities. Our evaluation (Section 4) reveals that LIME and SHAP score low in F5 (Interactivity) and F2.1 (Expressive Power), highlighting limitations in current XAI methods for user-adaptive, interactive explanations. The surveyed methods are largely non-interactive, providing static explanations with little to no flexibility for user interaction. As XAI advances toward greater interactivity and co-construction, new functionalities beyond current state-of-the-art methods are needed. To anticipate this shift, we have included metrics that may not yet be met by existing XAI methods and recognize the necessity for additional evaluation criteria in sXAI. We expect F5 (Interactivity) to evolve into a more detailed framework of its own. Given the demands of sXAI and recent advancements in neuro-symbolic AI [313], we foresee its growing role in XAI.

The F2 (Structure) property may require special attention in collaborative decision-making settings, where effective communication is crucial. Due to its user-dependent nature, objectively assessing structure is challenging, as perceptions vary based on domain expertise and cognitive preferences. Similarly, the F10: (Un)Certainty scale, currently defined with five levels, could be adapted to different use cases (e.g., simplified for general users while maintaining granularity for experts). Additionally, the F5.2 (Bias Detection) property may benefit from a structured user validation process to ensure that detected biases are both interpretable and actionable. Beyond these user-related factors, system-level considerations can also affect metric interpretation. In particular, the F11 (Speed) metric is sensitive to hardware configurations, implementation optimizations, and system load, factors that can affect explanation runtimes. We encourage developers and users to report these contextual and evaluation-related details along with their assessments, to support fairer comparisons, improved reproducibility, and more transparent benchmarking.

To remain relevant, the framework must evolve alongside emerging XAI trends, including social and collaborative systems. This entails redefining Interactivity (F5) to better capture user engagement and introducing new metrics to evaluate support for group decision-making. Additionally, regular updates integrating advancements such as generative AI [45] will help maintain its applicability in this rapidly changing field.

6 Conclusion

To foster the adoption of AI/ML systems in society and business, several essential AI principles must be meticulously considered and met by these systems. These principles encompass technical robustness, fairness, accountability, human agency, privacy, and transparency. XAI plays a pivotal role in addressing transparency by granting end-users the technical capabilities to anticipate the performance of AI/ML systems or when they approach critical scenarios. This article surveys a diverse array of XAI methods existing in the literature, acknowledging the challenge faced by AI/ML practitioners in determining the most suitable method for their specific applications.

Within this review, there is a detailed exploration of XAI methods with a particular focus on criteria, referred to as “properties”, that are frequently employed to evaluate the effectiveness of these methods. The systematic literature review reveals the absence of universal consensus on XAI assessment properties. Additionally, it highlights the limited formalization of “functionally-grounded” properties based on quantitative measures that do not necessitate explainee involvement. This gap complicates fair and objective comparisons among different XAI methods.

To bridge this research gap, this article advances the state-of-the-art by introducing a concise, consensual, and hierarchically structured framework for functionally-grounded properties. The framework is:

- Concise, as it consolidates overlapping properties identified in the literature.
- Consensual, as it builds on established formalizations where possible.
- Hierarchically structured, providing a clear and layered organization of properties (organizing 24 properties into 11 distinct property categories).

The framework’s practicality is demonstrated through an evaluation of LIME and SHAP, two widely used XAI methods, in a classification task. This study aims to help developers extract valuable insights within and across XAI methods, rather than identifying a universally superior method.

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